MAYA PATEL

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PROFESSIONAL SUMMARY

I build reliable backend services and practical developer tools for teams that need fast, observable software. Across internships and campus research, I have shipped event-driven APIs, data pipelines, and accessibility-focused web products used by students and operations teams. I enjoy turning ambiguous requirements into measurable engineering outcomes, from reducing batch-processing time to improving deployment confidence. I am currently completing my computer science degree at Georgia Tech and seeking software engineering internships and new-grad opportunities focused on backend, platform, or cloud infrastructure.

EDUCATION

Georgia Institute of Technology - Bachelor of Science - BS, Computer Science — Expected May 2027 | Atlanta, Georgia
Relevant Coursework: Data Structures, Algorithms, Computer Organization, Database Systems, Networking, Software Design, Linear Algebra, Probability

EXPERIENCE

Software Engineering Intern | HarborGrid Systems — May 2026 - August 2026 | Atlanta, GA

- Built a FastAPI service that normalized telemetry from 18 warehouse systems and processed 2.6 million daily events with idempotent retries.
- Introduced Redis caching and query batching, reducing dashboard API latency from 920 ms to 310 ms at the 95th percentile.
- Created GitHub Actions test and deployment workflows with preview environments, cutting average review-to-release time by 41%.
- Partnered with two product analysts to define reliability metrics and added Prometheus alerts that reduced undetected ingestion failures by 58%.

Undergraduate Research Assistant | Georgia Tech Accessible Computing Lab — January 2025 - Present | Atlanta, GA

- Developed a route-ranking pipeline for wheelchair-accessible campus navigation using sidewalk grade, curb-ramp, construction, and elevator data.
- Designed PostgreSQL/PostGIS schemas for 85,000 geospatial segments and accelerated route scoring 6.3x through spatial indexing and precomputed features.
- Ran usability sessions with 14 participants, converted findings into engineering tickets, and improved successful trip-planning completion from 64% to 86%.

Teaching Assistant - Object-Oriented Programming | College of Computing — August 2025 - May 2026 | Atlanta, GA

- Led weekly labs for 42 students covering Java, interfaces, inheritance, testing, and memory-aware data-structure design.
- Authored automated grading tests for six assignments and reduced manual regrade requests by 29% through clearer failure messages and rubrics.

SELECTED PROJECTS

AccessPath Campus Navigator / *Python, FastAPI, PostGIS, React, MapLibre* — February 2025 - Present

- Built an accessible route planner that compares distance, slope, curb-ramp coverage, indoor connectors, and temporary closures across 2,100 campus destinations.
- Implemented A* search with configurable accessibility penalties and returned most routes in under 180 ms after graph preprocessing.
- Added offline-friendly route snapshots and plain-language directions for users with intermittent connectivity.

CourseGraph Planner / *TypeScript, Next.js, PostgreSQL, Prisma, Docker* — September 2025 - January 2026

- Created a prerequisite-aware degree planner for 1,800 catalog courses with conflict detection, term availability, and graduation-path simulations.
- Designed constraint checks that identified missing prerequisites and impossible term combinations before schedule submission.
- Deployed containerized services with seeded demo data and integration tests covering 120 scheduling scenarios.

ADDITIONAL EXPERIENCE HIGHLIGHTS

- Worked across HarborGrid Systems, Georgia Tech Accessible Computing Lab, and College of Computing in roles requiring independent implementation, structured review, and clear status communication.
- Combined Python, Java, TypeScript with database, cloud, testing, and version-control workflows to move projects from initial requirements through documented validation.
- Balanced performance and reliability goals by measuring latency, throughput, data quality, or user completion before and after technical changes.
- Prepared design notes, test plans, demos, and handoff documentation so classmates, researchers, and internship teammates could reproduce results and maintain the work.

PROJECTS

TraceLite Observability SDK / *Java, OpenTelemetry, Gradle, JUnit* June 2025 - August 2025

- Published a lightweight Java instrumentation library for request timing, correlation IDs, error classification, and structured JSON logs.
- Benchmarked under synthetic load and kept median instrumentation overhead below 1.8% across 50,000 requests.

Realtime Study Board / *Node.js, WebSockets, React, Redis* March 2025 - May 2025

- Developed a collaborative whiteboard with presence, cursor updates, optimistic edits, and room-level permissions for student teams.
- Used Redis pub/sub to synchronize sessions across three app instances and supported 240 concurrent editors in load tests.

SELECTED TECHNICAL CONTRIBUTIONS

- Designed project architecture for AccessPath Campus Navigator and CourseGraph Planner, separating interface, application, data, and infrastructure concerns to support iterative changes.
- Implemented validation and error-handling paths for expected inputs, malformed records, partial failures, retries, and recovery scenarios rather than testing only the happy path.
- Used automated unit, integration, or hardware/system tests to verify core behavior and added repeatable fixtures or seeded datasets for regression testing.
- Profiled or benchmarked critical workflows and used measured results to prioritize indexing, caching, batching, concurrency, memory, or communication improvements.
- Created deployment or build automation with version control and documented environment setup, configuration, observability, and rollback expectations.
- Presented technical tradeoffs to peers and nontechnical stakeholders, connected implementation choices to user or operational outcomes, and incorporated review feedback.

LEADERSHIP & ACTIVITIES

- HackGT Organizer - coordinated a 36-hour student hackathon, mentor office hours, and sponsor technical workshops for 700 participants.
- Women in Computing Project Lead - mentored five first-year students building a campus food-availability dashboard.
- Open-source contributor - submitted documentation, test coverage, and accessibility fixes to three Python and React repositories.

TECHNICAL SKILLS

Languages: Python, Java, TypeScript/JavaScript, SQL, C, Bash

Backend & Data: FastAPI, Node.js, PostgreSQL, PostGIS, Redis, Prisma, Pandas

Frontend: React, Next.js, HTML/CSS, MapLibre

Cloud & Tools: AWS, Docker, GitHub Actions, Linux, Prometheus, OpenTelemetry, Git

Testing: Pytest, JUnit, Playwright, integration testing, load testing

ACADEMIC APPLICATIONS

- Applied Data Structures, Algorithms, and Computer Organization concepts when selecting data structures, algorithms, interfaces, and performance constraints.
- Used ideas from Database Systems and Networking to design storage models, data flows, communication boundaries, or experiment plans.
- Connected classroom work in Software Design to maintainable code organization, testing strategy, documentation, and team review practices.
- Completed technical writing and presentation deliverables explaining assumptions, limitations, evaluation methods, and next-step recommendations.
- Maintained reproducible repositories with setup instructions, sample data, issue tracking, and versioned changes suitable for independent review.

CERTIFICATIONS

- AWS Certified Cloud Practitioner - 2026
- Google Cloud Computing Foundations - 2025

HONORS & AWARDS

- Dean's List, four semesters
- HackGT Accessibility Track Winner, 2025

ADDITIONAL DETAILS

Professional Interests: production-quality systems, measurable performance, careful testing, accessible documentation, and cross-functional delivery.

Profile Status: Synthetic testing profile; no real employment, credential, or identity claim is represented.

Links: github.com/mayapatel-sample | www.linkedin.com/in/maya-patel-sample | mayapatel.example.com